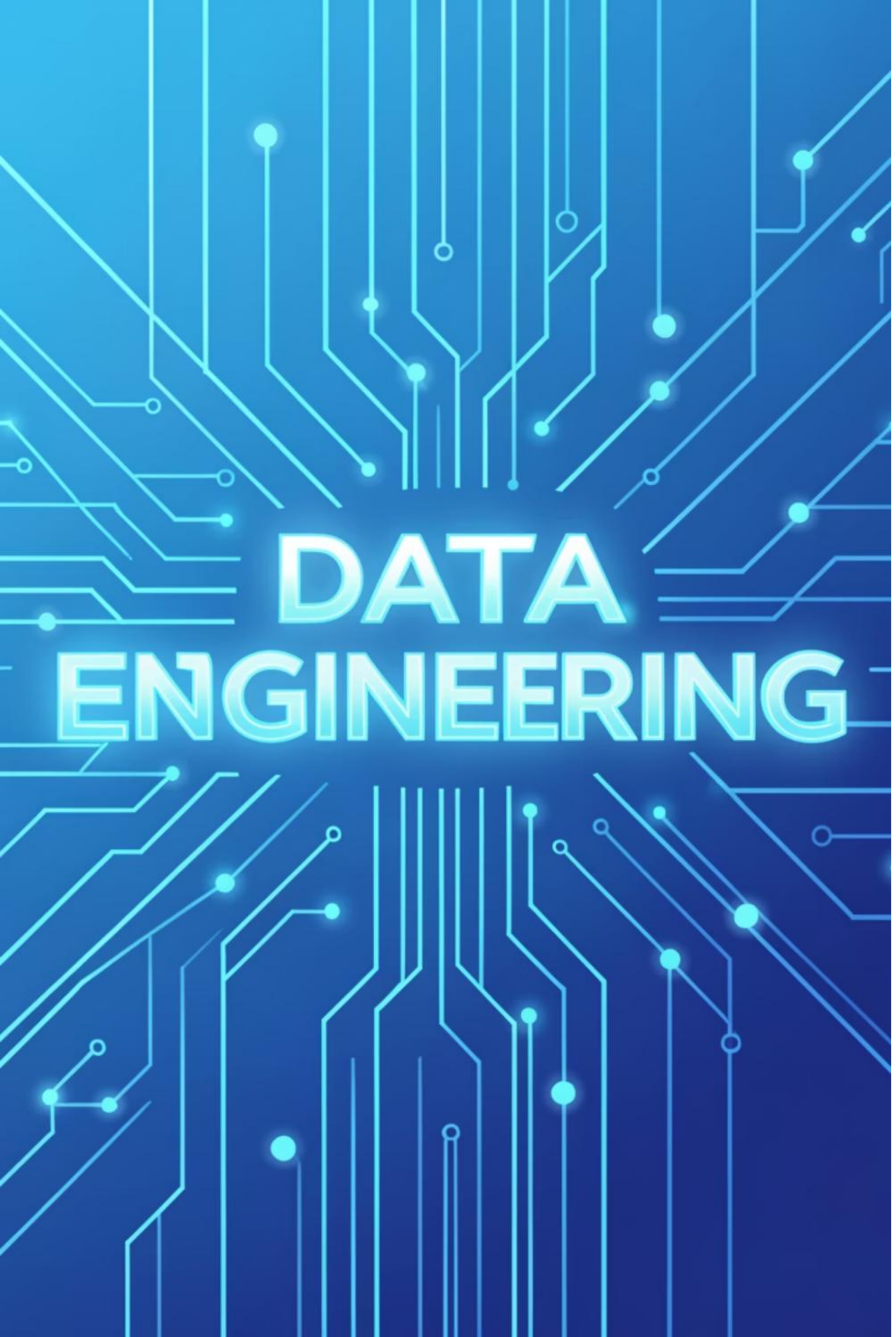


Data Engineering

Spring '26 – DS463

Feb 12, 2026

Lecture 13

Logistics: Your Course Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

Credit Hours

2 hours (Lectures) + 1 hour (Practical (Practical Session)

Office Hours

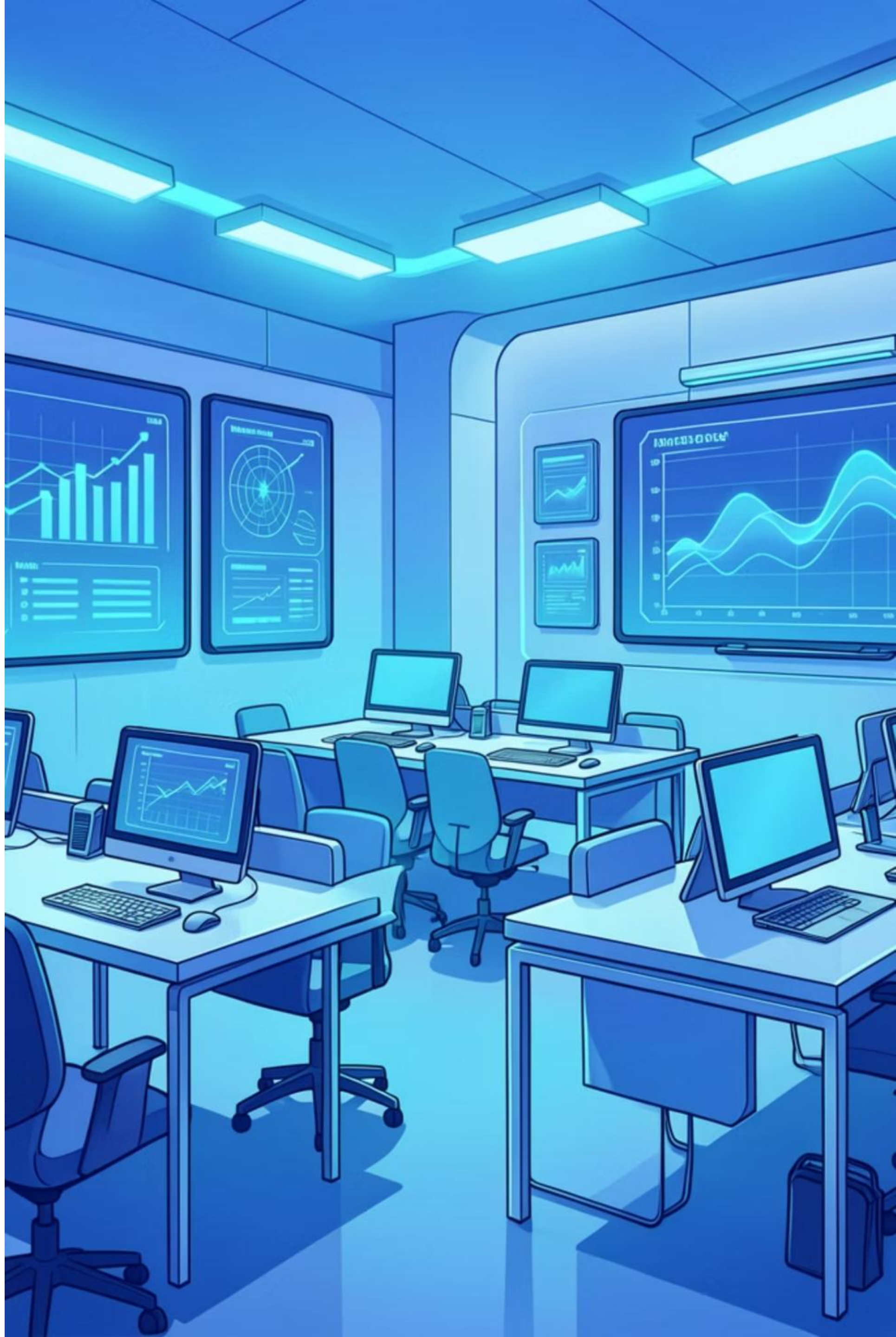
Monday, Tuesday: 10:00 AM – 11:00 AM, S18 2nd Floor NAB or by appointment.

Contact & Support

MS Teams: w5t93yk

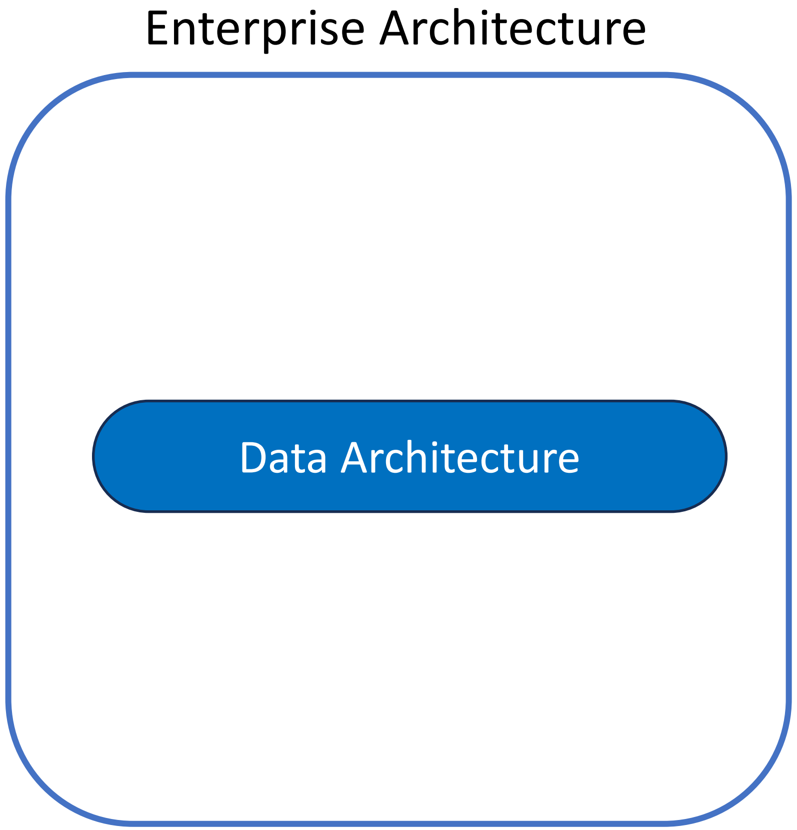
Email: farah.saeed@giki.edu.pk

WhatsApp: Reach out for quick queries



What is Data Architecture?

Enterprise Architecture



“The design of systems to **support change in an enterprise**, achieved by flexible and reversible decisions reached through careful evaluation of tradeoffs.”

“The design of systems to **support evolving data needs of an enterprise**, achieved by flexible and reversible decisions reached through careful evaluation of tradeoffs.”

Principles of Good Data Architecture

- Choose common components wisely
- Architecture is leadership

How data architecture impacts other teams and individuals

- Always be architecting
- Build loosely coupled systems
- Make reversible decisions

Data architecture is ongoing process

- Plan for failure
- Architect for scalability
- Prioritize security
- Embrace FinOps

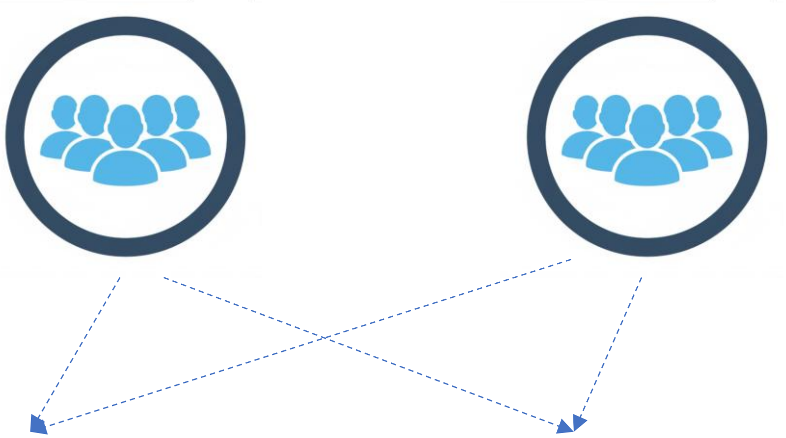
Unspoken but understood priorities

Principles of Good Data Architecture

- **Choose common components wisely**

How data architecture impacts other teams and individuals

Identifying tools and components that can benefit all teams



Common component   Common component

Examples
- Object Storage
- Version control
- Processing engines

Plan for Failure

Practical and Quantitative Approach

- System metrics

Availability

Reliability

Durability

Plan for Failure

Availability

The <u>percentage of time</u> an IT service or component is expected to be in <u>operable state</u>.

Amazon S3

Examples of S3 Classes	S3 One Zone-IA	S3 Standard
Availability	99.5%	99.99%

Plan for Failure

Availability

The <u>percentage of time</u> an IT service or component is expected to be in <u>operable state</u>.

Amazon S3

Examples of S3 Classes	S3 One Zone-IA	S3 Standard
Availability	99.5%	99.99%
Annual Downtime in hours	44-hour downtime	1-hour downtime

Plan for Failure

Reliability

The probability of a particular service or component performing its intended function during a particular time interval.

Plan for Failure

Durability

The ability of a storage system to withstand data loss due to hardware failure, software errors, or natural disasters.

Amazon S3

Durability

99.999999999%

(11 nines)

Plan for Failure

Recovery Time Objective RTO

The maximum acceptable time for a service or system outage.
For example: Consider the impact to customers.

Recovery Point Objective RPO

A definition of the acceptable state after recovery.
For example: Consider the maximum acceptable data loss.

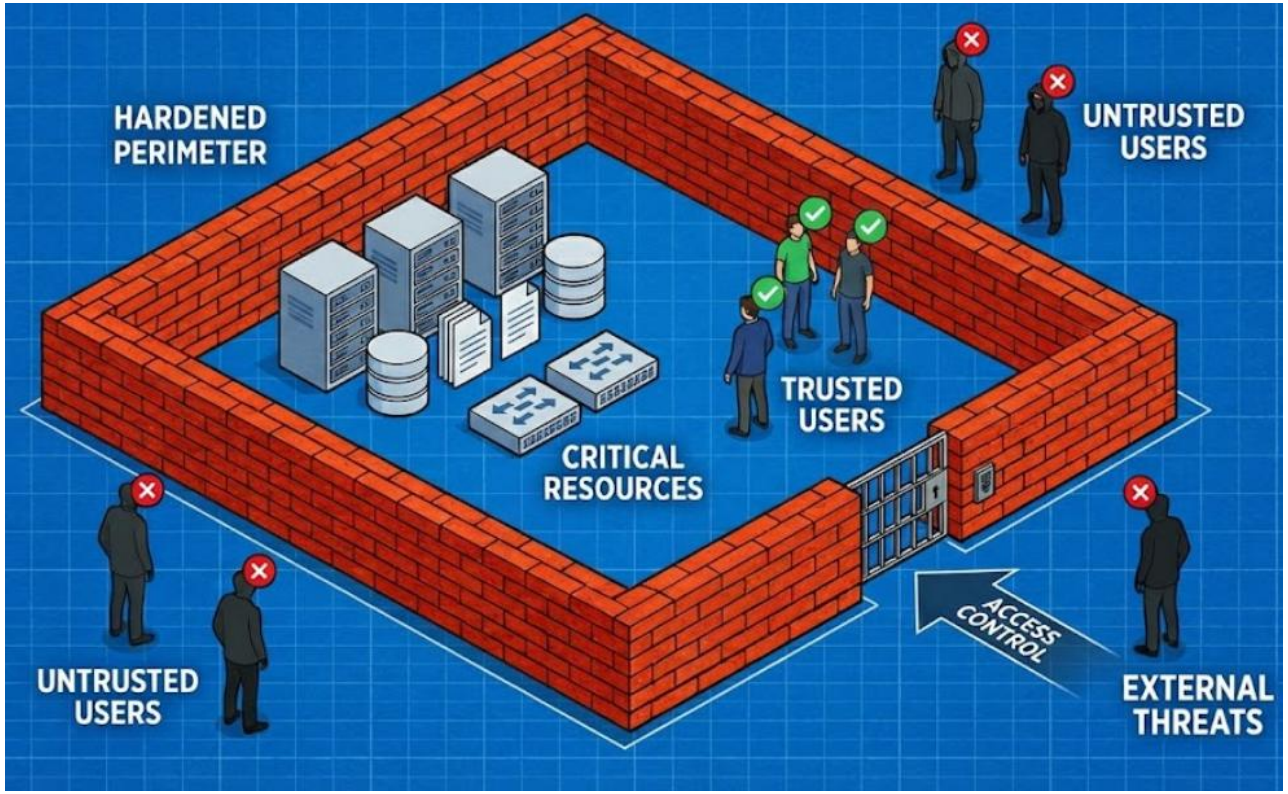
Prioritize Security

- Principle of least privilege

Prioritize Security

Hardened Perimeter Approach

An attacker only needs to get inside the wall



Prioritize Security

Zero trust security

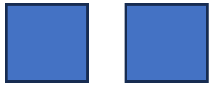
Do not trust both internal and external users

Data Management Approaches and Architectures

Feature	Data Warehouse	Data Lake	Data Mesh
Architecture	Centralized	Centralized	Decentralized (Distributed)
Data Type	Structured	All types	All types (domain-oriented)
Ownership	Central IT/Data Team	Central IT/Data Team	Domain-driven
Primary Use	BI & Reporting	Data Science/ML	Scalable, cross-functional analytics
Speed	Quick (for query)	Quick (for ingestion)	Quick (due to parallelization)

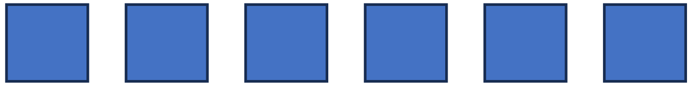
Batch Data Architecture

Batch Data Architecture



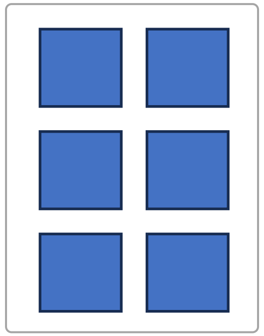
Batch Data Architecture

Batch Data Architecture



Continuous stream of events

Batch Data Architecture



Batch/ chunk

Batch Data Architecture